

Andrew Olvera

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RESEARCH INTERESTS

Agile, generalist control policies; reinforcement learning; generative world models. Current focus: scalable pretraining for control, using privileged classical controllers as teachers to generate synthetic supervision at scale and distilling it into general policies that transfer to vehicles and observations the teacher never had. Three first-author preprints in preparation.

EDUCATION

York College of Pennsylvania

Aug 2023 – May 2026

B.S. in Computer Science, York, PA

- Dean's List, Fall 2025 and Spring 2026.
- Coursework: Data Structures, Algorithms, Linear Algebra, Calculus, Probability & Statistics, Computer Vision, Computer Graphics.

Cabrini University

Aug 2021 – May 2023

Undergraduate coursework, Radnor, PA; transferred to York College

RESEARCH EXPERIENCE

Starscream: Pretrained Policies for Agile Drone Racing

2026 – Present

Independent research · andrew-olvera-research.github.io/site/starscream/

- Designed a synthetic course corpus that stratifies courses by the behaviors they demand (reversals, hairpins, dives, slaloms, braking), with layout-disjoint train and evaluation splits.
- Pretrained a **5.49M-parameter** causal transformer policy from supervision alone, with no RL, to isolate what representation learning buys before policy-gradient fine-tuning: an MPCC expert (acados) with DAgger relabeling, DART noise, expert-prefix roll-ins, and an auxiliary one-step dynamics head.
- Designed a gate-frame tokenization (six ordered route tokens plus state and control history; 130 Hz control, 11 ms action delay) and a test-time speed command that generalizes to unseen values and appears to select a behavior mode as well as a pace.
- Randomized the vehicle every episode (mass $\pm 15\%$, thrust map, rate limits, drag, wind and gusts, center-of-mass offset). Without being told the plant, the policy completes **60.6%** of attempts on 52 unseen layouts and **45.5%** on 92 unseen hard layouts, and turns back to recover missed gates.
- Distilling the privileged teacher into a vision-based student with camera gate-segmentation tokens and an auxiliary plant-inference head.

Tells: Opponent Inference in a MicroRTS World Action Model

2026 – Present

Independent research · andrew-olvera-research.github.io/site/microrts/

- Built a counterfactual MicroRTS corpus of **5.9M transitions** by cloning the engine and swapping the agent's action while holding the opponent's fixed, so the model learns causal action effects; episode-disjoint splits, held-out CoacAI evaluation.
- Found that cell accuracy was misleading (an early tokenizer scored 94.9% on cells but reproduced no frame exactly) and replaced it with an exact categorical temporal-JEPA tokenizer reconstructing **99.85%** of held-out frames.
- Trained a **103.1M-parameter** flow dynamics model (**0.973** one-step changed-cell F1), evaluated with action-shuffle gaps, counterfactual branches, and real games after finding that low-loss Dreamer-style models still produced failed policies.
- Distilled a 16-code opponent belief from fog-of-war history that improves long-horizon return prediction. PMPO policies trained entirely in imagination beat CoacAI in held-out panels; matched ablations showed the belief does not yet cause the wins, which I report as a result.

Learning Dynamics in RLVR Near Capacity

2025 – 2026

Independent research · andrew-olvera-research.github.io/site/token-level-rewards/

- Controlled two-phase study of RLVR on Qwen3-0.6B (GSM8K): fixed a stable baseline across GRPO vs. DAPO and group sizes 8/16/32, then varied only the reward (binary, structured, rank-based). Full-parameter fine-tuning on a single 16 GB GPU with TRL and vLLM.
- Showed that GRPO and DAPO both assign one advantage per response and differ only in per-token weighting, so within-group reward ordering is the only lever for finer credit.
- Rewards that separate completions within a group keep learning signal alive after binary rewards saturate, yet every configuration plateaus at **69–72%** from a 68% SFT start: at this scale RLVR sharpens the existing policy rather than adding capability.

PREPRINTS IN PREPARATION

A. Olvera. *Starscream: Scalable Pretraining for Agile Flight*. In preparation, 2026.

A. Olvera. *Tells: Inferring opponent intent in a MicroRTS world action model*. In preparation, 2026.

A. Olvera. *Learning dynamics in RLVR near capacity*. In preparation, 2026.

INDUSTRY EXPERIENCE

Sapio Sciences

May 2025 – Present

Application Engineer Intern, York, PA

- Develop on Sapio's lab informatics platform (LIMS/ELN), supporting biotech and pharma R&D customers from requirements through deployment.
- Build application-level configurations (data models, workflows, and integrations) that translate scientists' lab processes into working platform solutions.
- Contribute platform-level infrastructure changes and triage customer-reported issues down to root cause, working across support, product, and engineering teams.

Dataforma

May 2025 – Jan 2026

Software Engineer Intern, Web Development, York, PA

- Refactored legacy JSP modules into a Vue.js frontend and Kotlin backend, improving maintainability and enabling new feature development.
- Partnered with QA to debug defects, execute test plans, and validate deployments.

TECHNICAL SKILLS

ML & RL: PyTorch, Hugging Face Transformers, TRL, vLLM; imitation learning (Dagger, DART), teacher–student distillation, policy gradients (PPO, GRPO, DAPO, PMPO), RLVR, RLHF/DPO, model-based RL, world models (JEPA, flow matching), domain randomization, computer vision

Control & Simulation: Control theory, MPC/MPCC (acados), ROS, Flightmare, Isaac Sim, Basilisk, MATLAB/Simulink

Languages: Python, C++, C, Java, Kotlin, JavaScript, C#, SQL

Tools: Linux, Git, Docker, Bash, HDF5, WebDataset, Agents (Codex, Claude Code)